# MITESH ADAKE

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## EDUCATION

**University of Southern California** **Aug 2024 - May 2026**
*Master's, Computer Science* *Los Angeles, CA, USA*
- **GPA:** 3.75/ 4.0
- **Coursework:** Applied Natural Language Processing, Advanced Computer Vision , Adversarial and Trustworthy Foundation Models

**Pune Institute of Computer Technology** **Jul 2019 - May 2023**
*Bachelor's, Computer Engineering* *Pune, MH, India*
- **GPA:** 3.84/ 4.0
- **Coursework:** Data Structures & Algorithms, Machine Learning, Natural Language Processing, Deep Learning

## SKILLS

- **Programming Languages**: Python, Java
- **Databases / Query Languages**: MongoDB, MySQL, Oracle (SQL)
- **Cloud Platforms**: Amazon Web Services, Google Cloud Platform
- **AI / Machine Learning**: Neural Networks, Natural Language Processing, Large Language Models, PyTorch, Tensorflow, Keras
- **Tools & Frameworks**: Ab Initio, Dataiku, Docker, Flask, FastAPI, Django, Power BI

## PROFESSIONAL EXPERIENCE

### FedEx

*Sr. Revenue Science Analyst* *Plano, TX, USA* · **Jun 2026 - Present**
- Driving AI initiatives by integrating LLMs, ML models, and enterprise tools to improve pricing workflow and operational efficiency for Pricing Analysts and Sales teams

*Revenue Management - Intern* *Los Angeles, CA, USA* · **Sep 2025 - May 2026**
- Built 3 automated systems leveraging data engineering to automate pricing analyst workflows, cutting processing time from ~5 days to ~1 hour
- Partnered with cross-functional teams on PowerBI to create dashboards and AI proof-of-concepts for data-driven pricing systems

*Revenue Management - Summer Intern* *Memphis, TN, USA* · **Jun 2025 - Aug 2025**
- Designed and developed an automated system using Python, SQL, Streamlit, and Docker to process user inputs and perform complex, business-critical database operations
- Reduced manual processing time from ~10 days to ~10 minutes to optimize workflow logic, significantly increasing operational efficiency

*Software Engineer* *Hyderabad, TL, India* · **Aug 2023 - Jul 2024**
- Migrated Dataiku data pipeline to Ab Initio (ETL) data pipeline for Harmonized Code search engine, resulting in annual cost savings of $ 1M for FedEx and accelerating data processing time
- Executed an optimization strategy in Python for Natural Language Processing (NLP) code, significantly reducing data discrepancies from 25% to negligible levels and eliminating data shuffling from 75% of text data

### Persistent Systems

*Machine Learning Intern* *Pune, MH, India* · **Jun 2022 - Aug 2022**
- Implemented NLP-based question deduplication on company's community portal, improving efficiency and reducing duplicate postings to 10%
- Leveraged Python, Flask API, FAST API, and Docker to create a scalable and robust solution

### NICE

*Data Engineer Intern* *Pune, MH, India* · **Nov 2021 - Apr 2022**
- Built scalable, serverless real-time file ingestion using AWS (S3, DynamoDB, Lambda) and MongoDB for efficient file processing
- Streamlined data management tasks by implementing a user-friendly Python Flask interface

## PROJECTS & OUTSIDE EXPERIENCE

**Fast-Hack: GGUF k-quant Content-Injection Attack** Project Link **Jan 2026 - May 2026**
- Researched and built a new quantization attack for LLMs: stay benign under full-precision (FP) inference but become malicious after GGUF k-quantization
- Evaluated on Llama-3.2-1B-Instruct and Qwen2.5-1.5B-Instruct across Q2_K, Q3_K_M, Q4_K_M, Q5_K_M, and Q6_K with ASR up to ~82–83% on Q4_K_M with FP ASR ~0–3%; other quants showed ~38–68% quantized ASR with ~2–18% FP

**Hate Speech Unlearning for LLMs** **Jan 2025 - Mar 2025**
- Implemented Task Vector Arithmetic, Contrastive Learning, and Fine-tuning to remove hate speech from LLaMA 3.1 models, resulting in a drop of harmful rate from 62% to 34% on PKU-SafeRLHF dataset

**PriceNet: Stock Price Prediction using Large Language Models** **Sep 2024 - Nov 2024**
- Designed and implemented PriceNet, using Llama and Gemini for explainable financial time series forecasting with accuracy of 50% and ROGUE-2 of 0.546 on reasoning of predictions